Chapter 7 — Waves and Vibrations

Full textbook exercise · PECTAA Physics 11 · No ALP filtering

Q.1 — Multiple Choice Questions (with correct option and reason)

7.1. The simple wave speed equation is:

(a) v = fλ **(b)** v = λ/f
(c) v = f/λ **(d)** v = f + λ

Answer: (a) v = fλ — Speed = frequency × wavelength.

7.2. The principle of superposition states that the displacement of a wave is:

(a) the sum of displacements of its components **(b)** the product of components
(c) the difference of frequencies **(d)** the ratio of amplitudes

Answer: (a) the sum of displacements of its components — Displacements add vectorially.

7.3. A node in a stationary wave is:

(a) a point of maximum displacement **(b)** a point of intermediate displacement
(c) a point of zero displacement **(d)** a point of infinite displacement

Answer: (c) a point of zero displacement — Nodes remain at rest.

7.4. An antinode in a stationary wave is:

(a) a point of maximum displacement **(b)** a point of minimum displacement
(c) a point of zero displacement **(d)** a point of infinite displacement

Answer: (a) a point of maximum displacement — Antinodes oscillate most.

7.5. Stationary waves are defined as waves that:

(a) move with constant velocity **(b)** move with changing velocity
(c) oscillate in a fixed position **(d)** propagate through a medium

Answer: (c) oscillate in a fixed position — The pattern does not travel.

7.6. Harmonics are:

(a) integer multiples of a fundamental frequency **(b)** integer submultiples
(c) random frequencies **(d)** non-integer multiples

Answer: (a) integer multiples of a fundamental frequency — $f_n = nf_1$.

7.7. Constructive interference results in:

(a) a decrease in amplitude **(b)** an increase in amplitude
(c) no change in amplitude **(d)** a shift in phase

Answer: (b) an increase in amplitude — Waves reinforce.

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7.8. If amplitude is tripled, energy increases by:

(a) 3 times **(b)** 6 times
(c) 9 times **(d)** 12 times

Answer: (c) 9 times — Energy ∝ amplitude².

7.9. What type of waves do headphones use to produce sound?

(a) electromagnetic waves **(b)** mechanical waves only
(c) pressure waves only **(d)** longitudinal waves

Answer: (d) longitudinal waves — Sound in air is longitudinal.

7.10. The typical frequency range of microwaves is:

(a) 10^2–10^4 Hz **(b)** 10^9–10^{11} Hz
(c) 10^{14}–10^{16} Hz **(d)** 10^6–10^8 Hz

Answer: (b) 10^9–10^{11} Hz — Microwaves are in the GHz range.

7.11. The bending of waves around an obstacle is called:

(a) refraction **(b)** reflection
(c) diffraction **(d)** interference

Answer: (c) diffraction — Diffraction spreads waves.

7.12. The Doppler effect is used in astronomy for:

(a) measuring diameters of stars **(b)** determining velocity of galaxies
(c) analyzing black holes only **(d)** studying electromagnetic waves only

Answer: (b) determining velocity of galaxies — Red/blue shift gives radial velocity.

Q.2 — Short Answer Questions

7.1. What are the conditions for interference to occur?

The sources must be coherent, have the same frequency and a constant phase difference, and the waves must overlap with comparable amplitudes.

7.2. Differentiate between constructive and destructive interference.

Constructive interference occurs when waves arrive in phase, path difference nλ, giving maximum amplitude. Destructive interference occurs when they arrive out of phase, path difference (2n+1)λ/2, giving minimum amplitude.

Feature	Constructive	Destructive
Phase	In phase	Out of phase
Path difference	nλ	(2n+1)λ/2
Result	Maximum amplitude	Minimum amplitude

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7.3. What are coherent waves and coherent sources? Give examples.

Coherent waves have the same frequency and a constant phase difference. Coherent sources maintain this relation, such as two loudspeakers driven by the same oscillator or light from a laser split into two beams.

7.4. Distinguish between longitudinal and transverse waves.

In longitudinal waves particles vibrate parallel to propagation, as in sound. In transverse waves particles vibrate perpendicular to propagation, as in light and string waves.

Feature	Longitudinal	Transverse
Vibration	Parallel	Perpendicular
Example	Sound	Light
Polarization	Not possible	Possible

7.5. Can two identical waves travelling in the same direction form a stationary wave?

No. Stationary waves require two identical waves travelling in opposite directions. Same-direction waves simply superpose to form a travelling wave.

7.6. How is the Doppler effect applied in studying cardiac problems?

Ultrasound reflected from moving blood cells changes frequency. The shift gives blood flow speed and direction, helping detect blockages or valve problems.

7.7. What is diffraction of waves? For what purpose is a ripple tank used?

Diffraction is the spreading of waves around obstacles or through gaps. A ripple tank shows water-wave diffraction and interference visually.

Constructed Response Questions

7.1 Which measurement of a wave is most important for determining intensity?

Amplitude, because intensity is proportional to the square of amplitude.

7.2 Can the Doppler effect be applied to light waves?

Yes. Light from a receding source is red-shifted and from an approaching source blue-shifted. The shift gives the relative speed along the line of sight.

7.3 Compare compressions and rarefactions of a longitudinal wave with crests and troughs of a transverse wave.

Compressions are regions of high pressure/density, like crests of maximum displacement; rarefactions are low pressure/density regions, like troughs. Both represent the periodic extremes of the wave.

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7.4 How should a source move so that the observed frequency does not change? Give two examples.

It should have no radial motion relative to the observer, for example moving in a circle around the observer or moving perpendicular to the line joining source and observer.

7.5 Why is it difficult to recognize beats when the frequency difference is greater than 10 Hz?

The loudness variations become too rapid for the ear to follow as separate beats; they are heard as a rough or continuous tone.

Comprehensive / Long Questions

7.1 State and explain the principle of superposition. Apply it to noise-cancelling headphones.

When waves overlap, the resultant displacement is the vector sum of individual displacements. Noise-cancelling headphones produce a wave equal in amplitude but opposite in phase to the noise, so destructive interference reduces the sound.

7.2 What are standing waves? Illustrate an experiment demonstrating standing waves on a stretched string.

Standing waves form when two identical opposite waves superpose, producing nodes and antinodes. A string fixed at one end and driven by a vibrator at the other shows stationary loops at resonant frequencies f_n = nv/2ℓ.

7.3 Find the frequencies of harmonics in an organ pipe open at both ends and closed at one end.

Open pipe: f_n = nv/2ℓ, all harmonics. Closed pipe: f_n = nv/4ℓ with n odd, only odd harmonics. The closed pipe fundamental is half that of an open pipe of the same length.

7.4 Define and exemplify diffraction. Describe a ripple tank experiment.

Diffraction is spreading of waves through gaps or around obstacles. In a ripple tank, plane water waves pass through a narrow gap and spread into circular ripples, showing diffraction most strongly when the gap is comparable with the wavelength.

7.5 What are beats? Prove beats per second equal the difference of frequencies.

Superposing two waves of frequencies f_1 and f_2 gives a resultant whose amplitude varies at frequency $|f_1 - f_2|$. Thus the number of loudness maxima per second equals the frequency difference.

7.6 What are progressive waves? Discuss intensity of progressive waves.

Progressive waves travel through a medium transferring energy. Intensity is power per unit area, I = P/A, and is proportional to amplitude squared.

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7.7 Keeping the Doppler effect in mind, analyze (a) source moving away (b) source moving toward a stationary observer.

(a) Wavefronts are spread out, wavelength increases and observed frequency decreases: $f' = vf/(v + u_s)$. (b) Wavefronts are crowded, wavelength decreases and frequency increases: $f' = vf/(v - u_s)$.

Numerical Problems (fully worked)

7.1 Wave speed on a string is 24 m s⁻¹. Length 6.0 m. Find fundamental driving frequency.

Given: $v = 24\ m\ s^{-1}$, $\ell = 6.0\ m$

Formula: $f_1 = v/2\ell$

$f_1 = 24/(2\times6) = 2$ Hz.

Answer: 2 Hz

7.2 Lowest resonance frequency of a guitar string length 0.75 m is 400 Hz. Find wave speed.

Given: $\ell = 0.75$ m, $f_1 = 400$ Hz

Formula: $v = 2\ell f_1$

$v = 2\times0.75\times400 = 600\ m\ s^{-1}$.

Answer: 600 m s⁻¹

7.3 Fork A produces 4 beats/s with B. Loading B with wax increases beats to 6/s. If A is 320 Hz, find B when loaded.

Given: $f_A = 320$ Hz, initial beats = 4, loaded beats = 6

Formula: $f_{beat} = |f_A - f_B|$

B could be 316 or 324 Hz. Wax lowers frequency; only 316 Hz can increase the difference to 6 Hz.

Loaded B = 320 − 6 = 314 Hz.

Answer: 314 Hz

7.4 A steel wire supports 80 N, diameter 0.50 mm, length 1.5 m, density 7.8 × 10³ kg m⁻³. Find fundamental frequency.

Given: $F = 80$ N, $d = 0.50$ mm, $\ell = 1.5$ m, $\rho = 7800\ kg\ m^{-3}$

Formula: $\mu = \rho\pi d^2/4$, $v = \sqrt{(F/\mu)}$, $f_1 = v/2\ell$

$\mu \approx 1.53 \times 10^{-3}\ kg\ m^{-1}$.

$v \approx 229\ m\ s^{-1}$.

$f_1 \approx 229/3 \approx 76.2$ Hz.

Answer: 76.2 Hz

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7.5 Sunlight intensity 500 W m⁻² falls on a 0.50 m² panel for 4 h. Find energy.

Given: I = 500 W m^{-2}, A = 0.50 m², t = 14400 s

Formula: E = IAt

E = 500 × 0.50 × 14400 = 3.6 × 10^6 J.

Answer: 3.6 × 10^6 J

7.6 (a) I = 16 W m⁻², amplitude 2 m; find k. (b) I = 25 W m⁻², k = 5 W m⁻⁴; find amplitude.

Given: $I = kA^2$

Formula: $k = I/A^2$, $A = \sqrt{(I/k)}$

(a) k = 16/4 = 4 W m^{-4}. (b) $A = \sqrt{(25/5)} = \sqrt{5} \approx 2.24$ m.

Answer: (a) 4 W m^{-4} (b) 2.24 m

7.7 (a) Sound power 200 W over 150 m²; find intensity. (b) Light power 100 W over 400 m²; find intensity.

Given: P = 200 W, A = 150 m²; P = 100 W, A = 400 m²

Formula: I = P/A

(a) I = 200/150 ≈ 1.33 W m^{-2}. (b) I = 100/400 = 0.25 W m^{-2}.

Answer: (a) 1.33 W m^{-2} (b) 0.25 W m^{-2}

7.8 A radio antenna broadcasts 500 W. Find intensity at 10 km.

Given: P = 500 W, r = 10^4 m

Formula: $I = P/(4\pi r^2)$

$I = 500/(4\pi \times 10^8) \approx 4 \times 10^{-7}$ W m^{-2}.

Answer: 4 × 10^{-7} W m^{-2}

7.9 An organ pipe length 1 m. Find fundamental and first two harmonics (a) open (b) closed. v = 340 m s^{-1}.

Given: ℓ = 1 m, v = 340 m s^{-1}

Formula: open: $f_n = nv/2\ell$; closed: $f_n = nv/4\ell$, n odd

(a) 170 Hz, 340 Hz, 510 Hz. (b) 85 Hz, 255 Hz, 425 Hz.

Answer: (a) 170, 340, 510 Hz (b) 85, 255, 425 Hz

7.10 A train approaches at 90 km h^{-1} sounding 1000 Hz. Find apparent frequency approaching and receding. v_{sound} = 340 m s^{-1}.

Given: u_s = 25 m s^{-1}, f = 1000 Hz, v = 340 m s^{-1}

Formula: $f' = vf/(v \mp u_s)$

Approaching: $f' = 340 \times 1000/(340-25) \approx 1079.4$ Hz.

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Receding: $f' = 340 \times 1000/(340+25) \approx 931.5$ Hz.

Answer: 1079.4 Hz, 931.5 Hz

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